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Decision Calculus of Protesters and Regimes: Eastern Europe 1989

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In our analysis of mass protest in the nonreformist part of Eastern Europe 1989 we ask not only why people rebel, but how they rebel against highly repressive regimes. We show how the decisions of individual protesters concerning participation in protest activity are influenced by a set of assurance games that involve their own and other social groups. We also show how regime actors go through the reverse process of a "deassurance game" as their instruments of coercion are undermined.

In the fall of 1989, the world was astounded by popular mass protests bringing down the Communist regimes of Eastern Europe. While it is understandable that outside observers were caught by surprise, the more intriguing issue is that the most engaged actors on both sides of the political spectrum within the Eastern European states were themselves unprepared for the phenomenon of effective mass protest. When protests began in East Germany and Czechoslovakia in October 1989, neither the dissidents nor regimes expected that they would produce fundamental political change within a few short weeks. Both sides recognized that the tremendous repressive powers of the government were still in place. Just as in the People's Republic of China, these governments had the abilities to physically attack demonstrators and to punish them afterward by denying them jobs, promotions, and other goods, and they were in command of these abilities as late as the summer of 1989. We need to explain how individual citizens made the decision to demonstrate against their powerful governments, we need to explain why the regimes failed to suppress the demonstrations, and we must explain why the process occurred so rapidly and thoroughly.

Arguably, most theories of rebellious behavior would have failed to predict the outcome. Resource mobilization theories would have pointed to the lack of organizational preconditions for effective protest. Rational Choice theories would have raised the free-rider problem, and relative deprivation theories would have had a difficult time explaining why mass protests started at this particular time and why they recurred with growing numbers day after day.

Tarrow's (1989, 1991) argument that protest has to be analyzed within a broader political opportunity structure would suggest focusing on the consequences of the

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reformist approach of Gorbachev as well as events in Poland and Hungary. Yet these exogenous influences alone cannot explain the phenomenon of collective action and its role in bringing down the resistant regimes of Czechoslovakia, East Germany, Bulgaria, and Romania. The question we try to answer has been well expressed by Olson (1990, 16): "How can autocratic regimes that appear to have such awesome power over their citizens collapse so quickly?"

When popular rebellion started in the hard-line states of Eastern Europe, Hungary and Poland had already embarked on a reform path, and we deal with them only to the extent that their experience influenced events elsewhere. Clearly there are numerous commonalities in the process of radical political change experienced everywhere in Eastern Europe, and the underlying causes for change are similar (Bunce and Chong 1990; Schopflin 1990), but we do not focus on these broader questions. Reform in Poland and Hungary and the reformist pronouncements of Gorbachev suggested that the power of repression in all communist regimes had become limited (Bunce and Chong 1990), and we shall refer to this demonstration effect. Yet this did not mean that protesters could rule out the possibility of repression and even harsh repression. As recently as in 1981, martial law had unexpectedly been introduced in Poland, and the Chinese regime cracked down brutally on protesters in June 1989. Honecker ordered to take a similar harsh approach in East Germany on October 9, 1989, and was stopped only at the last moment by his security chief.

Our purpose is to understand the mechanism and process of revolt against powerful and resistant regimes. These processes involve two distinct but interactive dynamics: the increasingly large and frequent demonstrations of the masses, and the simultaneous process of fragmentation, defection, and loss of confidence among the regime. We believe that the key to understanding these processes and their interrelation lies in recognizing the heterogeneity of both the "masses" and the "regime." The masses consist of dissidents, students, workers, and party supporters. Individual members of each of these groups face different constraints and will engage in protest activity under different conditions. We can only understand how the "masses" overcome their collective action problem if we disaggregate them and find out how the separate parts interact. We rely on the game-theoretic device of the assurance game to describe this process. We posit that two assurance games—one played by the individual and society as a whole, and another played by the individual and his or her social group—provide three signals that allow the individual to gauge whether "protesting" or "not protesting" is the best play.

If the protesters are playing an assurance game, then the individuals composing the regime are playing a "de-assurance game," looking for signals when to defect from their commitment to support the regime and its repressive actions. As the mass is heterogeneous, so is the regime. Part of the regime is the elite, but there are also civil officials as well as the security and armed forces. If these groups become "de-assured," the elite is no longer able to count on their repressive powers and is forced to make concessions or abdicate power.

We conceptualize rebellion as a series of events involving calculations and

recalculations by protesters and regimes. While Tarrow (1991, 13) is right that "protest is, first and foremost, a form of action," its other prime characteristic is that the actors calculate what others will do and what their own best move is in this light. Perceptions matter, because prediction about how others will act is at the core of decision making. We show how this process works for protesters as well as regime actors during the sequence of events that make up the "collapse" of Communism in the hard-line states of Eastern Europe. In using a research design that examines links between a sequence of events, we take account of past critiques of existing theories of rebellious action (Snyder 1976; Likens and Kohfeld 1983), and we try to explain how this dynamic process works in detail. After outlining the process itself we discuss complications, in particular how protesters overcame the free-rider problem and why repression failed so thoroughly.

Protest as an Assurance Game

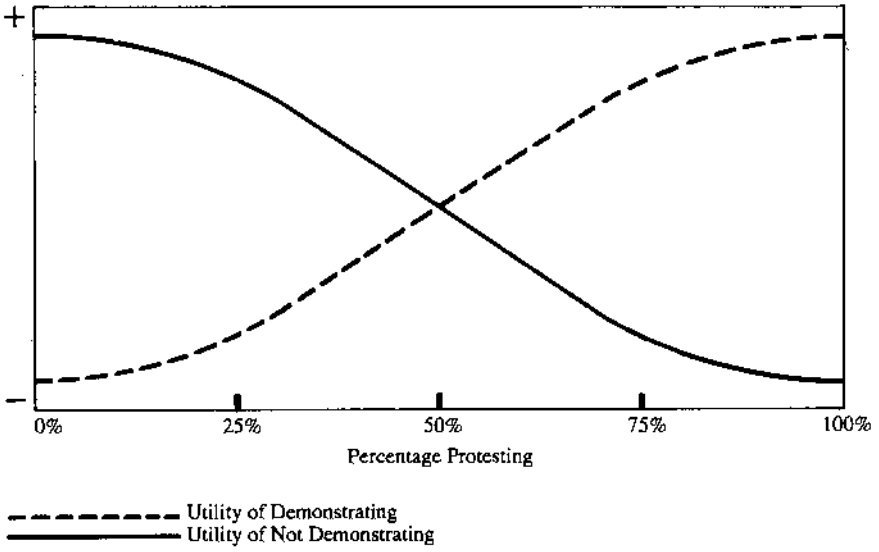
To understand the mechanics of mass resistance we must turn to two considerations in the calculus of the potential protester: protection and prediction. There are two aspects of "protection." First, there is protection against physical repression at the site of the demonstration. The overall size of the demonstration is crucial here. Secondly, there is the fear of being sanctioned in one's local environment, for instance at the university or the workplace. In this context, the number of protesters within the local environment is the key factor. For example, a worker employed at a certain factory might skip work to join a demonstration of 10,000 students. He might be protected by the size of the crowd from risking suffering a beating, but if he is the only worker from that factory participating then the management of the plant can single him out for job sanctions. The situation would be different if a significant number of workers at the same plant participated in the protest, because the same type of sanctions against this larger group would damage the operation of the entire factory. The same logic holds for university students and others.

The second element in the protester's calculus concerns prediction. Again, there are two elements that must be discussed. First, each potential protester wishes to know the chances of achieving protection at the protest site as well as in their group environment. Secondly, all potential protesters will wish to know the probability of regime collapse. They would prefer to participate in activity that might possibly bring about some meaningful change and are looking for clues and assurances that other social groups may join the protests and that the regime is weakening.

The key to understanding the timing and speed of the revolt in the hard-line states of Eastern Europe lies in the relationship between protection and prediction of various social groups. When certain groups achieve protection, individuals within other groups recalculate probabilities. Later, we discuss the interrelationship between protection and prediction for four social groups within the "mass": dissenters, students, workers, and party supporters.

FIGURE 1

SIMPLE *N*-PERSON ASSURANCE GAME



In trying to explain the decisions of individual protesters, we employ some of the conceptual logic and devices of game theory. The strength of game theory is that it assumes actors are strategic—the individual's action is based on how he or she thinks others will act. This assumption fits the mass demonstration phenomenon, especially concerning its "protection" aspects. Each individual may have wanted to vent his or her dissatisfaction with the regime for some time, but lone individuals are unlikely to go to the center of town and protest unless they believe a considerable number of other individuals are likely to do the same. Potential protesters look for assurances against repression within their smaller and larger environments. Theories of rebellion that focus too much on the individual's psychic state tend to ignore this.[1]

The game theoretic device most appropriate to analyze protest in Eastern Europe is the *n*-person assurance game, or tipping game (Sen 1967). Schelling (1985) has illustrated and drawn implications from the *n*-person assurance game, and we reproduce a "Schelling diagram" for our case. The lines on the graph represent relative values for the two choices of "demonstrate" and "not demonstrate." The only thing being considered here is the absolute difference in value between the two options for a given individual. The key point is that the individual pay-off for a choice is determined by the choices of the rest of the population. Figure 1

[1]Political diffusion models (Bartholomew 1982) also fail to capture the reality of these events. These models, which are similar to models explaining the spread of epidemics, are appropriate when applied

indicates that at very low levels of dissent, the individual value of "not demonstrate" is far superior to the decision to "demonstrate." The regime can easily identify and punish these dissenters. As the number of protesters increase, the relative power of the regime sanctions for any one individual decreases. The intersection of the two lines represents a highly unstable equilibrium where the value of protesting and not protesting is equal. After the intersection, "demonstrating" becomes optimal and movement toward a stable equilibrium at 100% can be expected. This represents a bandwagoning effect where no one wishes to be seen as supportive of a discredited regime without a future.[2]

The *n*-person assurance game is meant to capture processes with a sudden switch of choices in a binary decision. The crucial issue to be explained is how the percentage of protesters moves past the tipping point. Of course, the bandwagon effect will be set off more easily when the tipping point is lowered. This is accomplished when the value of "demonstrating" is raised, the value of "not demonstrating" is lowered, or when both occur simultaneously. There are several factors which move the lines in the Schelling diagram in these directions. Three are most relevant in our discussion of the demonstration process in East Europe: prediction, social motivation, and logistical costs.

First of all, the value of the Schelling diagram lines are in large part a representation of the individual's subjective beliefs about the regime's intentions to use violent repression. This is where the prediction element comes into play. Any event which brings about a reevaluation in which the regime is seen as less willing or less able to bring force into play will lower the tipping point. Secondly, any increase in social motivation to protest will raise the value of the "demonstrate" line and thus lower the tipping point. Thirdly, a decrease in the logistical costs of demonstrating will have similar effects; when information about the time and location of rallies becomes readily available and individuals can get to the event easily, the cost of the "demonstrate" option is reduced.

In Eastern Europe, these three factors interacted to reduce tipping points for large segments of the population and catalyze the snowballing protests of 1989. The first element, prediction, is incorporated within a generalized "model" developed immediately below; the other two are integrated into a latter section dealing with variation within this "model."

General Dynamics of the Demonstration Process

The simple "tipping game" mentioned earlier has a powerful intuitive logic and, on the surface, it may seem particularly appropriate for analyzing the East European cases. However, it does not capture the complexity of the potential protester's decision calculus. The major problem is that the Schelling diagram "tipping

to situations when the nature of the flow of information is critical to the question at hand. In the demonstrations considered here, the residents of the cities where the protests occurred all had similar information (see section on informal assembling mechanisms).

[2]For an excellent discussion of Schelling diagrams and tipping processes, see Margolis (1991).

point" may represent calculations of an average member of society, but in reality there is no such thing. Society is heterogeneous, not homogeneous as the Schelling diagram assumes. Secondly, the assurance game as described provides little basis to discuss the needs for prediction sought by potential protesters. There is basically only one signal for the individual to use to gauge the direction of the protest process (the overall percentage of those protesting) and this is too simplistic.[3]

In trying to explain the nature of the "snowballing" demonstration process witnessed in Eastern Europe, we outline a "model" that builds upon that work of Schelling. It is based on the following three insights.

Heterogeneity of Society

The timing of individual decisions to join the protests is largely a function of membership in certain social groups. In theory, the nature of social groups getting involved in protest can vary, but we focus on dissidents, students, workers, and party supporters. Communist regimes try to homogenize society by undermining traditional or newly forming group structures, but nevertheless groups can be defined by their social context and identity.[4] Universities and enterprises provide a daily interaction context for students and workers, and interaction on a regular basis is one core definition of social group (Truman 1958). Another basis for our definition of groups involves their level of opposition or support for the regime, as well as the regime's repressive policy. Dissidents and party supporters define the two poles on the continuum of support for the regime, with students and workers being in between. Loose group identities form on the basis of people thinking alike and communicating with each other. The groups are also defined by repressive policies of the regime that differ in quality and intensity between subgroups such as students and workers (Karklins). What's more, repression in Communist systems relies on peers informing on each other, or denouncing each other publicly (Kuran 1991, 28–29). This too reinforces informal or work-related group structures.

Dual Assurance Games

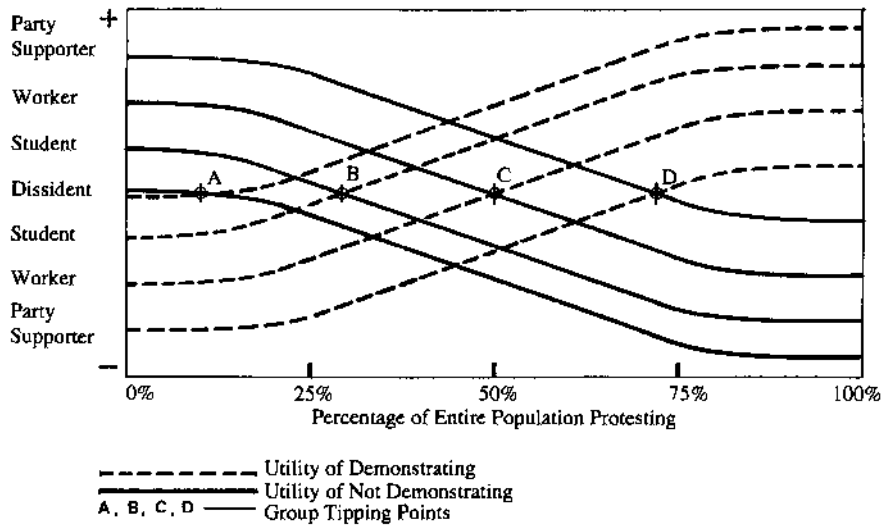
The assurance game nicely illustrates the individual's desire for protection against regime sanctions; however, as there are two distinct types of repression, there must be two types of corresponding assurance games.

[3]For a similar, but distinct, approach compare Granovetter (1978). Elster (1990, 3–6) proposes a process that is similar to ours, but sees it as stemming from differences in personality types more than from differences derived from the individual's membership in a social group.

[4]Heterogeneity within each group must be taken into account as well. Individual propensities to protest vary due to differences in age, education, sex, family responsibilities, and personal psychological make-up (Marsh 1977, Muller 1980). Thus, some individuals within each group will need more assurance than their fellows. Thus, bolder students may be joining the protest at the same time as more timid members of the dissenter group and younger workers will be joining at the same time as the more constrained students. Each social group has a "mean" with certain deviations; the activity of each group thus will overlap to an extent.

FIGURE 2

HYPOTHETICAL TIPPING POINTS FOR VARIOUS SOCIAL GROUPS



To avoid physical repression or arrest at the demonstration site, as well as to simply avoid being personally identified as a dissenter, the individual plays an assurance game with the general population. Even here we must be careful to incorporate an element of heterogeneity: different social groups are subject to different regime sanctions and have different motivations. As pointed out by Karklins (1987) those who have been engaged in dissent before and have already been sanctioned have less to lose than others willing to demonstrate more quickly. Students will usually have more to lose—they may jeopardize their future careers by participating. Thus, their tipping point will be somewhat higher. Workers' tipping point will be higher yet since they can be fired from their jobs. Party supporters will not begin demonstrating until they believe that they will have to deal with a new order. This belief will be in part related to a very high overall tipping point.

Figure 2 shows a set of hypothetical values and corresponding tipping points for various social groups in the Eastern European demonstration process.

The assurance game of the individual versus the general population addresses fear concerning general repression. The individual is also concerned about protection in his or her smaller environment, and thus a second assurance game is played with fellow members of a social group. When 100,000 out of 136,000 students participated in the general strike in Prague on November 27, 1989, the university administration could impose no sanctions against this percentage of their student body. A tipping point had been passed and protesting became the thing to do even

for apathetic students. After this, the Czechoslovak workers held discussions and voted whether they would join the protest. In these cases, tipping points for each plant were reached in one shot. The selective incentives involved with federated structure and small-n solutions (Olson 1965) are applicable in these cases.

Interaction of Tipping Points

The tipping point is lowered when expectations of violent regime repression are reduced. The question is how protesters make these predications. As will be discussed in a later section, the basis for protesters' predictions can come from outside the country. Certainly, Gorbachev's actions and the events in Poland and Hungary are crucial in understanding the demonstration process in Czechoslovakia and East Germany. However, by focusing so much attention on these exogenous influences analysts have overlooked signals found within the demonstration process itself. We hypothesize that when one social group has realized its tipping point and achieved a satisfactory protection level, other groups use this as a reference in predicting the effectiveness or willingness of the regime to use repression against their own group.

Logically, social groups will use the group preceding themselves in the tipping order as a referent. Thus, students look at dissenters, workers to students, party supporters to workers. The achievement of a tipping point by a referent group affects both the overall societal assurance game and the internal group assurance game of the following group. For instance, once students have passed their group tipping point workers can be pretty certain that students will continue to attend the protests in at least the same numbers. They can more easily predict that the size of the following rallies may increase to a level that surpasses their own threshold for personal safety. Concerning the internal game, the regime's failure to deter or stop the students produces the perception that it will do likewise in the case of workers. Individual workers and factory units will begin acting upon their recalculation of regime strength and the probabilities of regime collapse. Their action will push percentage cooperation in both the overall assurance game and the workers group assurance game closer to the relevant tipping points, eventually surpassing them.

This process is especially critical for the next group in the tipping order, the party supporters. After the workers, the supposed backbone of the Communist party, have reached a tipping point and protest without fear or repression, the party supporters must recalculate probabilities. Thus, the protest process can take on a life of its own by linking the reaching of protection by one group with another group recalculating predictions.

As the dissenter group has no group preceding it in the tipping order, it tends to make predictions based on other criteria. As mentioned, in the case of Czechoslovakia and East Germany one can also argue that protests in neighboring states formed a referent. The dissidents were the group most clearly attuned to the significance of the accelerating success of reforms in Poland and Hungary and the

creeping repudiation of the Brezhnev Doctrine. In this sense exogenous influences set off the general endogenous process described earlier.

Summary

Our model of protester behavior holds that the potential demonstrator is operating not according to one tipping point, as in the original Schelling diagram, but actually is using three such tipping points: the overall societal tipping point of their group, the internal tipping point within their own group, and the tipping point in a reference group. With the *n*-person assurance game, the crucial issue is raising the percentage of protesters past the tipping point. In our "model" we show two ways in which this happens. First, as seen in the societal game represented by figure 2, there is a "building block" process in which social groups with lower tipping points combine to push the percentage of participation toward the higher thresholds of other social groups. Secondly, within the group games, tipping points reached by reference groups serve to lower perceptions of regime will or effectiveness for groups following in the tipping order. Thus, group tipping points in the latter groups are adjusted downward.

While this model describes the actual calculations of protesters in our cases quite well, we use it mostly as a heuristic device how a chain reaction of protest can develop.[5] It shows that heterogeneity of actors is the key in the solution of the assurance game and in the depiction of any "snowballing" process. In the presence of abundant information, we must explain differential rates of joining protest activity. The evidence in Eastern Europe indicates that differences in timing of participation were linked to social group membership.

"Deassurance" and Regime Groups

Thus far, we have outlined a dynamic which explains how tens of thousands of individuals decide to demonstrate despite possible sanctions, now we need to connect this phenomenon to the actual collapse of the regimes. Repressive regimes do not fall simply because citizen protests illustrate the regime's lack of legitimacy. Again, we have to show the actual process of change and the connection between the "mass" and the "regime."

As Olson (1990) recently has pointed out, authoritarian regimes and dictatorships are inherently fragile as they rest upon the perceptions of their guards, officials, and administrators. If these perceptions suddenly change, the officials will cease to carry out their orders: "If the cadre observe a moment of vacillation, an incident of impotence, a division of leadership, or even a collapse of analogous regimes, all the power of an imposing regime can vanish in the night air" (Olson 1990, 17). We agree, but would emphasize that perceptions of regime officials may

[5]Certainly, there would be advantages to formalizing this "model." However, at this point in time, we find it difficult to accurately weight the importance of the two assurance games. Also, there are important complicating factors (see the "Two Puzzles" section in this article).

be changing the most dramatically during the demonstration process itself. We hypothesize that the same dynamic behind the actions of individuals who compose the "mass" can be applied to the decision process for the individuals who compose the regime: both have needs for protection and prediction related to various assurance games played out versus the society as a whole, within one's own social group and within referent groups.

As with the "mass," it is important to recognize the heterogeneity of the regime. At least four groups make up the repressive apparatus of the regime: civil regime officials (including managers and the media), regular police and the army, the security police, and the regime elite. Their choice of actions is not so clearly binary as with the "mass," but it is possible to speak about options of "repress" versus "don't repress" for each of these groups. For the media, the choice may be rephrased as "censor" or "don't censor," for soldiers, perhaps "shoot" or "don't shoot."

The important point is that individuals within these groups operate under different sets of constraints and will consider defecting from their support of the regime at different junctures of the protest process. We argue that the timing of defection is related to signals emanating from various assurance games.

The logic of the internal group assurance game is similar to the one for protesters. Journalists and officials wishing to defect will feel protected from possible sanctions if a significant percentage of their own group is about to defect or have already done so. A majority of journalists or factory managers cannot be fired without resultant high costs. Soldiers and police are involved in a similar calculus: "not shooting" while the majority plays its "shoot" option will involve court-martial or other severe sanction as the regime will probably survive under this contingency. On the other hand, "shooting" while others opt for "don't shoot" will entail equally severe sanctions imposed by a likely new regime.

The case of the soldiers emphasizes the need for prediction by the individuals in the regime. Soldiers must try to predict the solution to their internal game. At the same time, the security police are doing likewise, although their position is different from that of the regular army and police. They may not be able to avoid sanctions under a new regime even by playing "don't shoot." Regime administrators too are trying to predict whether the regime will survive. Once doubts arise they have little motivation to support regime repression, especially harsh repression, since the latter might close the door to governmental employment in the future. The more visible elements of the regime, journalists for instance, will be striving to predict the chances for regime change at the earliest possible moment because future job prospects will be brightest for those who appear as leaders rather than those who are simply joining the "bandwagon."

The basis for these predictions, or changes in perception, probably lie in the same cues for action used by the demonstrators. When the society as a whole has reached the tipping point and is demonstrating day after day without individual fear of physical sanctions, regime officials upgrade the probabilities of regime collapse. When the social groups that have joined the protest process include not

only students but workers, soldiers will recalculate the chances of other soldiers firing upon the crowd. In turn, the regime elite will need to recalculate its ability to rely on the soldiers under these new conditions.

Thus, we see two processes simultaneously. On the one hand, social groups of the mass are becoming assured of protection versus sanction and upgrading the possibilities of regime change. Simultaneously, groups within the regime are using signals emanating from the mass to become "deassured" of the regime's ability to survive. Groups within the mass will be using referent groups to gauge the direction of the movement with an order of dissidents, students, workers, and party supporters. In the regime a similar order may be found with officials "tipping" first, followed by army and police, security police, and finally the regime elite itself.

Data on regime groups are insufficient to assess the empirical fit of our theory in detail. In the segments that follow, we therefore focus on the protester side and refer to "the regime" as if it were an unified actor.

Two Puzzles

Thinking in terms of an assurance and deassurance game among a heterogeneity of actors helps to understand how the mass protests in Eastern Europe could escalate, but it ignores two major puzzles. The first puzzle concerns the calculations of the individual protester in the early stages of the demonstration process. Although we do not have a classic example of a collection action problem here,[6] the free-rider dilemma for the potential demonstrator during low percentages of protest is the same: collectively it is best if everyone protests, but individually it is best not to protest. Olson's (1965) solutions to this free-rider problem do not seem to apply in these cases: neither economic selective incentives nor small-n solutions (federal structure) have convincing widespread applicability. The relevance of other solutions (Moore 1989) is also questionable.

The second puzzle concerns the failure of regime repression. First, why was prophylactic repression so ineffective, and secondly, how can one explain that the numbers of protesters increased dramatically *after* the use of brutal repression. In 1989, the regime in East Germany used large-scale police brutality against demonstrators on October 7 and 8, and faced a huge rally on October 9; in Czechoslovakia, the regime employed brutal police action on November 17 and faced huge rallies by November 20; in Romania, the Ceausescu regime slaughtered civilians on December 17 and watched a revolution break out on December 21. Why did brutal police action not only fail to act as a deterrent in these cases but instead actually acted as a catalyst?

These two puzzles bring up the issue of variation. After all, in many other cases the collective action problem is not solved, regime repression is effective, and the

[6]The situation in Eastern Europe is best portrayed in game theoretic terms as an n-person assurance game, whereas, a collective action problem is generally pictured as an n-person prisoner's dilemma. The key difference being that in the latter it is always optimal to defect while in the former, it is optimal to defect at lower levels of cooperation but not at higher levels of cooperation.

general bandwagoning effect never begins, is stopped, or is reversed (the case of Tiananmen will be discussed later as an example). In the previous section, the "model" developed was based almost entirely upon the heterogeneity of domestic social groups and the interactive effects produced by their presence at rallies during the protest process. There are other factors, though, outside of the numbers and categories of protestors that come into play to change the nature and direction of the assurance games played by the various social groups. The answer to the aforementioned puzzles and an explanation for variation within cases can largely be found in the influence of four factors: public goods motivation, informal assembling mechanisms, demonstration effects, and focal events.

All of these factors raise the value of the "demonstrate" line or lower the value of "not demonstrate" line on the Schelling diagram. They produce variation through their presence or absence, the timing of their appearance within the general demonstration process, and their separateness or conjunction. When several of these factors come together at one time, tipping points can be reduced dramatically in a fashion with which the regime is unable to cope. On the other hand, the bandwagoning process can be stopped or reversed when these factors do not coalesce, when the regime takes steps to mitigate their influence, or when the protesters fail to employ a strategy that utilizes advantages. An understanding of these four factors is necessary to comprehend the variation in the demonstration processes as they were played out in Czechoslovakia, East Germany, and China—the case studies here.

Public Goods Motivation

The first answer to the free-rider problem and the failure of regime deterrence concerns the existence of public goods motivations. This is a specific type of social motivation which raises the value of "demonstrate" and helps lower tipping points to more reachable percentages. Rational Choice models assume that people actually perceive that their individual contributions have no real impact upon the probability that the good will be realized. Much empirical work shows that this is simply not the case. As Muller and Opp (1986) show "in contrast to conventional rational choice models of rebellion, the public goods variables, not selective incentives, appear to be the most important incentives for performing rebellious political action" (483). Clearly, some in Eastern Europe were acting according to a form of what Muller and Opp (1986) term "collective rationality" in which the realization that collective action will fail if there are too many nonparticipants motivates individuals to participate under the belief that their individual contribution will be meaningful. In Eastern Europe, this phenomenon was captured in the commonly heard phrase, "If not me, then who?"[7]

We agree with Muller and Opp that the possibility of social goods resulting

[7]Perhaps Elster (1989) has described this phenomenon the best with his moniker for such "irrational" actors of "everyday Kantians"—people who ask the very question cited earlier.

from collective action can be a motivating force, but we go further than they by identifying the point in the series of mass protests where this factor has the most political meaning. Specifically, it motivates dissidents and students more than any other social groups and it does so at times of rising numbers of protesters. This factor does not rise automatically from the hearts of publicly minded individuals but instead arises in the context of the overall strategic situation—it is strongest when individuals first come to believe in the efficacy of mass collective action. In Eastern Europe, this belief was also initially strengthened by the experience of reform in neighboring states.

Additionally, the public goods factor may help produce changes in the overall societal assurance game. As groups with lower tipping points achieve their safety levels and come out to the streets to protest, the chances of regime collapse, or at least significant change, are raised in the minds of potential protesters. As Muller and Opp (1986) have shown in their research, social motivation is higher when chances of success are higher. There is increased willingness to take risks involved with appearing at a demonstration. Thus, the passing of tipping points by social groups earlier in the tipping order produces higher social motivation in other groups. The end result is that the tipping points seen in Figure 2 move leftward making the initiation of a bandwagoning process more likely.

Informal Assembling Processes

Mass protests have an organizational side: Both common sense and resource mobilization arguments about rebellion suggest that someone has to coordinate the time, place, and specific purpose of protest. High costs of participation in terms of gathering information will lower the "demonstrate" line and raise the tipping point to higher percentages. As the regime controlled the official media, it would have seemed that it could have used this advantage to significantly change the payoffs in the assurance game in its own favor. Considering that many protests took place, the job for protesters seems even larger. How did dissident groups manage to accomplish these tasks in spite of the authorities' year-long efforts to preclude precisely such activity?

A close look at events in Eastern Europe in the fall of 1989 reveals that dissident groups had a limited role in organizing protests and they took on a leadership role only in the last phase when someone was needed to negotiate with the authorities. Protest mostly resulted from a number of informal assembling mechanisms and social learning processes.

The timing of the initial demonstrations was determined by anniversary dates such as the 40th anniversary of the German Democratic Republic (October 7), the 71st anniversary of the founding of Czechoslovakia (October 28), and the 50th anniversary of the slaying of a student demonstrator by German occupying forces in Prague. Later on, informal slogans were generated at each demonstration about their continuation such as "meet every Monday" in Leipzig and "meet every day" in Prague after November 17. The place of the demonstrations was often

predetermined by the existence of traditional assembling places such as central city squares. Other assembling places were set by precedent, for example, East Germans meeting in and around churches.

Informal social ties were also to assist the assembling process. Intellectuals, students, and later workers, communicated with each other by word of mouth at their institutions and places of work. There is also evidence that information was passed on through informal second economy networks where people knew and trusted each other due to the unofficial exchange of goods and services.

When demonstrations started, the official media inadvertently publicized coming events by issuing warnings about the consequences of unauthorized protest. In later stages, when the authorities began to emphasize dialogue, the official media provided direct coverage of events. The informal nature of the assembling mechanisms helps explain miscalculations by the authorities. Their knowledge of the limited organizational means of dissident groups apparently led them to believe that mass protest was impossible. Just as the regimes were unaware of the potential of mass participation in protest evolving through group-centered solutions to the assurance game, they were unaware of the power of mass self-organization once a chain reaction of protests was underway.

Demonstration Effects

In addition to all this, a demonstration effect between the states within the East European system helped overcome the free-rider problem and mitigated the deterrent value of the regimes' repressive powers. In the previous "model," the achievement of internal tipping points of reference social groups served to reduce the perception of the regime as a willing and effective repressor. Of course, in Eastern Europe, this same effect was also accomplished by external events. The internal tipping process described earlier was embedded within an East European regional tipping order. Citizens of East Germany were well aware of the changes that took place in Poland and Hungary and thus knew that Communist regimes were capable of losing their monopoly of power. The flight of thousands of East Germans across the Hungarian border, with the tacit consent of the Hungarian government, furthermore showed that each state would have to deal with its problems on its own. Even more so since the Soviet Unions' gradual repudiation of the Brezhnev Doctrine suggested that outside military intervention as the ultimate repressive means was precluded. As East Germany encountered massive opposition, Czechoslovakian citizens further recalculated the value of regime repression. This logic can be carried further for the Bulgarians and Romanians, but of course, the regimes in these two cases recalculated and chose new strategies: the Bulgarian regime choosing to attempt cooptation by way of earlier concessions and the Romanian regime choosing open repression at an earlier stage.

Demonstration effects were most decisive in the initial stage of protest. First, they helped overcome the free-rider problem which is most intense when the group of protesters is small. While it is always individually nonrational for an

individual to demonstrate, it is even more so when few others are about to do so. By October 19, the precedent of successful social protest in Hungary and Poland, as well as the success of East Germans fleeing to West Germany helped to assure the initial protesters that they would not remain alone for long. Even though this did not solve the free-rider problem altogether, it mitigated it.

Furthermore, information about reform and protest in neighboring states created a demonstration effect changing the parameters for the effectiveness of both deterrent and physical repression. Yet, while subgroups of protesters were quick to pick up on these cues, the East German and Czechoslovak regimes were slow to do so and were soon doomed by applying the wrong measures at the wrong time.

The Failure of the Repressive Calculus of the Regimes

Examining the evidence from hindsight, the regimes of East Germany and Czechoslovakia seemed to possess the following logic concerning what actions to take at different levels of overall protest (the Bulgarian and Romanian regimes updated their own codes apparently based on the information received from the East German and Czechoslovakian cases): At very low levels of protest, use prophylactic repression (control dissident organizational capacities and isolate dissident leaders in order to decapitate potential protest movements); do not use open repression because its deterrent effect is much more powerful if kept in reserve as the uncertainty concerning its use can be a as powerful as its actual use. If prophylactic repression fails, use open but limited repression. If protest continues to grow, use open repression before things get out of hand. If demonstrations reach and maintain a level so large that open repression seems infeasible, initiate minor concessions. If protest is unaffected by minor concessions, grant major concessions.

On the surface, such a strategy seems eminently reasonable. However, it failed because it assumed that repression has a constant meaning when in fact the effectiveness of repression varies. In our cases the effectiveness of repression was undermined at two crucial times: prophylactic repression failed in the initial stage due to demonstration effects as well as informal assembling mechanisms and public goods motivation of the first groups of protesters. A conjunction of these three factors brought a dramatic reduction in dissident and student group tipping points. Once intense physical repression was used in the third stage of the protests, it created a "focal event" and backfired.

The effectiveness of the regime's strategy of repression was first undercut when the reform precedent in fraternal states encouraged the initial group of protesters to express themselves, and thus triggered the assurance game. It appears that the regimes' only chance of staying in power would have been to skip its first action outlined in table 1 and immediately proceed to the next stage of open repression. Since the group of people protesting at this stage was limited, the use of arrests and physical violence might still have prevented the spread of protest. If not, the regime could have tried to apply its third-stage option of large-scale repression

TABLE 1

Stage	Regime Action	Group Action	Outcome
1st Stage very low level of protest; outside shock	Prophylactic Repression	Dissenter tipping point reached; some student action	Low levels of protest reached in overall terms
2nd Stage of Protest	Open but Limited Repression	Student tipping point approached	Moderate levels reached
3rd Stage	Large-scale Repression	Student tipping point reached; first worker defections	Overall tipping point is reached but situation is still unstable
4th Stage	Minor Concessions	Movement of workers grows; first Party defections	Levels of protest stabilize at number clearly past overall tipping point
5th Stage	Major Concessions	Workers reach tipping point; Party defections continue	Massive levels of protest reached

along the lines of the crackdowns in China in June 1989 and Poland in December 1981. As it was, the East German and Czechoslovakian regimes never used such extreme measures and applied their harshest repression at a time when its main effect was to trigger more protest.

Focal Events

Surveys of the theoretical and empirical literature on the nexus between dissent and coercion (Eckstein 1980; Lichbach 1987; Zimmermann 1980) show it to be complex. The correlation between protest and government coercion is not necessarily either direct or inverse. It can fit various curves, and the direction of causation may vary. The crucial point for our discussion is that "coercion may escalate or de-escalate protest" (Karklins 1987, 321).

At certain stages in the development of mass protest, acts of violence on the part of the regime become "focal events," that is, events which become the focus of further protest in and of themselves and intensify protest by highlighting the lack of legitimacy of the regime. The importance of a focal event depends upon its timing and the nature of the information it transmits to potential protesters. Focal events are thus connected to both social motivation and a reevaluation of the regime's repressive capabilities. Thus, focal events are in themselves a combination of forces that affect the tipping point.

In our cases, both the timing and the informational content of the regime's use of violence were of a crucial nature. The timing of regime violence came when students were consolidating their protest and before many workers were involved.

The violence served as an issue that linked the students and workers. Before the regime stepped up its repression in Czechoslovakia and East Germany, these two groups were divided by social and economic differences, but once the regime used violence neither group could justify it, and thus some workers joined students in antirepression rallies that would soon become antiregime rallies.

The strength of the link produced between different social groups by the focal event depends on the outrage produced by that event. The higher the anger and feelings of solidarity with the victims, the more the different groups will put aside their differences and focus attention against the violence itself.

Moreover, apologies and indecisiveness in the aftermath of the regime violence in Czechoslovakia and East Germany informed the population that the regime was not as strong-willed as had been thought. Also, the belief that the Soviets and other Warsaw Pact nations would not interfere was driven home at this point. Gorbachev had stated as early as July 6 that sovereignty should not be limited and the use of force between Warsaw Pact members must be ruled out; however, actions speak louder than words and the fact that the regime used considerable violence without any support, material or verbal, by the Soviets must have convinced wavering individuals that there would be no repeat of 1956, 1968, or 1981.

The focal events in Czechoslovakia and East Germany played a major role in the speedy progress of the assurance game. The regime employed violence before the decisive tipping point—that of the workers—was reached, and the violence, contrary to the expectations of deterrence theory and rational choice theory in general, worked to thrust the percentage of those demonstrating past that point. The use of violence should have shown the will and the ability of the regime to use its powers, in terms of figure 1 it should have raised the value of the "don't demonstrate" line and actually pushed the tipping point farther to the right. In reality, the opposite occurred. In other words, rather that acting as a deterrent, violence acted as the catalyst for further protest.

Summary of East European Mass Protest Process

Both the protesters and the regime are basically rational players, but boundedly rational in the sense that they are operating according to certain rules of thumbs about how to act.[8] In the absence of complete information, this is the best players can do. The important point is, though, that players are indeed calculating according to strategic cues that serve both as guarantees against sanctions and promises of a different future.

Table 1 illustrates how the decision rules of both the regime and the various social groups interacted to form a rapid, but hardly spontaneous, snowballing demonstration process. To show the accuracy of this scheme, we turn to a case study of Czechoslovakia; comments on East Germany and China follow.

[8]In more game theoretic terms, the actors here might be labeled "boundedly Bayesian" in that they are updating information under incomplete information.

The Czechoslovak Case: Calculations

The assumptions mentioned earlier about the strategic thinking of different social groups and the regime can be seen most easily in the Czechoslovak case. We proceed by stages according to table 1.[9]

The first stage in Czechoslovakia lasted between January and October of 1989. Here we see how a dissident community moves toward an internal tipping point in spite of the continuing effectiveness of regime repression. In January, hundreds of dissenters engaged in week-long demonstrations to mark the 20-year anniversary of Jan Palach's self-immolation in protest of the forcible suppression of the Prague spring. When the police arrested activists and sentenced Vaclav Havel and others to prison, the prophylactic repressive powers of the regime were effective in stopping open protest for the time being. However, within the dissenter community, the regime repression served as a focal event: Havel's show trial and the accompanying media campaign gave dissenters an issue to rally around. New independent groups formed at increasing rates and 33,000 people signed a petition protesting against persecution of dissidents.

Developments in neighboring Communist states also contributed to the activization of the dissidents. The renewed legalization of Solidarity and its meteoric rise to power in the summer of 1989, as well as the emphasis on reform in Hungary, the Soviet Union, and (by October) East Germany, could not have helped but bolster the belief in Czechoslovakia that continued dissident activity could bring lasting change.

The second stage of protest was reached in Prague on October 28 when for the first time thousands of students joined committed dissidents in an unauthorized rally commemorating the seventy-first anniversary of Czechoslovak independence. This demonstration made clear that prophylactic repression had failed to stop open protest in spite of the intensive efforts by the regime. Before the demonstration, the official press published numerous warnings and many dissenters were arrested under false pretense, detained, or ordered out of Prague. Some people were scared off, but between ten thousand and twenty thousand others joined in the biggest demonstrations since 1969. Once the demonstration was under way, the police arrested 355 and used some brutality, yet open repression was still limited relative to that seen in the next stage.

When students planned a rally for November 17 to mark the killing of a student protester by German occupation forces in 1939, the Czechoslovak regime first tried to coopt the demonstration by authorizing it. However, when nearly fifty thousand students turned the event into a pro-democracy rally, police riot troops engaged in unprecedented violence that brought the entire process onto a new plane. The brutality at this event became a focal point for the immediate following demonstrations and the general strike called for November 27.

[9]Data on the events in Eastern Europe in 1989 and 1990 were compiled from numerous newspaper and magazine reports as well as from the useful publications of Radio Free Europe Research, especially their weekly *Background Reports* and *Weekly Record of Events* 1989, and the weekly *Report on Eastern Europe* 1990.

This case shows how police brutality, coming at a crucial stage in the unfolding protest process, can become a "focal event" that triggers a rapid escalation of anti-regime demonstrations. Brutal repression increases anger and feelings of solidarity with the demonstrators who became victims, but in addition, the demand for rectifying the injustice becomes the leading trigger of the next demonstration. And precisely because demonstrators make past repression the issue, the regime is put at a disadvantage in the repeated use of repression. The delegitimization of future government violence is especially effective if the protesters are clearly nonviolent and have limited and popular demands. This is the crucial argument made by the advocates of nonviolent civil disobedience who focus on turning the political effect of government repression against the government itself. Such thinking did exist among East European dissidents who since the crushing of the Hungarian uprising in 1956 had rejected violence and other extreme measures, and had increasingly emphasized that their actions were legal (Bugajski and Pollock 1989, 69).

The focal nature of the regime's violence, however, did not reach its full backfiring effect until several days after its occurrence. First, the fear of massive physical repression had to be overcome. In a three-day period, November 18–20, we see a remarkable compacted example of the signaling and snowballing processes involved in a series of mass demonstrations. On November 18, the day after the crackdown, 2,000 individuals returned to the scene of the violence to lay flowers in commemoration. On November 19, about 20,000 individuals, mostly students, demonstrated against the regime action. On November 20, the figure was 200,000 and for the first time it appears that some workers had become active (there were only 136,000 students in Prague). The levels reached half a million by November 25. We can see the strategic thinking involved at this juncture—as people saw that the Communist regime was paralyzed, they quickly joined the demonstrations. The 2,000 on November 18 (perhaps dissidents or student activists) signaled to the rest of the students that the regime effort at deterrence was not going to be entirely effective. The 20,000 of November 19 (mostly students) signaled to the rest of the students and the population at large that the regime had no will to continue the violence and that the Soviets would not be intervening. The 200,000 of November 20, showed that the tipping point had been reached within the student group and that some workers were beginning to become active for the first time in 20 years: the possibility of social tipping point being reached, one that would force a change of the regime, became a real possibility that activated the public goods motivations of many potential protesters.

At this point the students and dissidents were unified, but their actions did not necessarily mean that the regime was in its death throes. The crucial issue until November 27, the day of the general strike, concerned the decisions of the workers. Both students and the party focused on this critical reference group which had always been given preferential treatment by the Communist regime. In the words of a student and a strike organizer, "We won't win if the workers are not on our side. To the leaders of our country, we students are nothing. But workers are the power here. To persuade them is critical to our cause" (Gwertzman, 240). The

workers also saw themselves as the critical group. Deliberations occurred at most factories. As a worker pointed out about discussions at her plant:

> This was not spontaneous support. We talked about what would happen if the strike didn't work, what kind of repression there might be. But we decided we can't support a government that beats its students. Next time it will be us, and then who will strike? (Gwertzman & Kaufman 1990, 242)

This last quote nicely illustrates many of the key aspects of the protesters in Eastern Europe. First, they calculated their actions and did not act spontaneously; second, they took the threat of repression very seriously; third, their calculations about future possibilities were made by examining regime actions against a reference social group; finally, regime violence acted as a focal event to unite social groups.

At this point, the fourth stage had begun and the regime began to offer minor concessions to defuse the demonstration process before it could become consolidated at an unmanageable point. Older, hard-line ministers were replaced and promises to punish and reign in the security forces were given. Instead of exhibiting strength and the possibilities of compromise, though, regime concessions were seen as a sign of the fragility of the regime—a sign that continued pressure would bring the regime down. The general strike of November 27 demonstrated the failure of the regime strategy at this fourth stage: millions of people participated, 900 factories and businesses were closed with only hospitals, nursing homes, and a few other enterprises continuing to function.

Indeed, workers had clearly followed students. The party, having to face up to this, employed its last desperate strategy—major concessions. By this time, however, many party members realized that the probabilities of the regime remaining in power were not very good. Two of the strategic clues, a high overall tipping point, and the tipping point of a crucial reference group—the traditionally allied workers—made the end of the regime foreseeable. The party began to unravel and disintegrate beginning a movement toward its own internal tipping point. Even hard-liners and true believers were forced to evaluate their future under a new order.

East Germany: Regime Miscalculation

Both protesters and regimes try to anticipate the outcome of the next event, as well as subsequent ones. We have tried to explicate how these calculations are made, and what assumptions are made. One important aspect of this dynamic evident in Eastern Europe is that immediately preceding events are used to anticipate the outcome of the next event. Protesters and regime groups use reference groups and the change in the overall level of protest to gauge how they should act. Although this may be a fully rational process given the constraints of incomplete information, it can easily lead to miscalculation.

Miscalculation occurs because in a dynamic process, actors are updating the information in their own set of calculations while basing their assumptions about others' action on the preceding play; yet the lessons learned from the preceding

event may not apply in next event. For instance, a set of potential protesters may see the inabilities of the regime in using limited forms of open repression and draw the conclusion that the regime is unable to use repression effectively; thus, they may demonstrate believing that doing so is harmless. At the same time, the regime may be calculating that because their limited repression failed it will be necessary to use large-scale repression the next time.

In Czechoslovakia, this type of regime miscalculation occurred in the transition from the second to the third stage of protest when the regime calculated that harsher repression was the best way to deter future demonstrators. While this may have been effective at an earlier stage, the demonstrators already had a different decision calculus. Many more people were participating due to the cues they had received from the students (who were reaching an internal tipping point), and from the regime, which had been using only limited police action.

The dynamic of regime miscalculation is even clearer in East Germany. The stages of protest and regime action outlined in table 1 capture events well, both in terms of the demonstrations and the specific form of East German protest by flight to West Germany (due to limits of scope, the paper will not depict this process here). Between September 25 and October 3, during the early stages of small demonstrations in Leipzig and East Berlin, the regime relied mostly on its power of preemptive coercion and was restrained in the use of actual punishment. It used police repression more openly and intensely on October 4 when thousands demonstrated in the railway station in Dresden while trains from Prague transporting East German refugees to West Germany passed through. The third stage was reached on October 7 and October 8 when thousands demonstrated in several cities and were met with police brutality and arrests. Again, outrage over police brutality served as a focal event that catalyzed further protest, as evidenced by the large demonstration in Leipzig on October 9. The societal tipping point for participation in protest was reached and the usefulness of repression disappeared. According to many reports, the hard-line East German party boss, Honecker, initially wanted to accelerate police repression on October 9 by issuing a "shoot to kill" order but was blocked by the security chief who told him, "we can't beat up hundreds of thousands of people" (Gwertzman, 219).

Thus began the fourth stage of the East German rebellion as the regime tried to contain protest with limited changes in personnel as well as gestures of engaging in dialogue with the protestors. Nevertheless, over the next few weeks, thousands participated in repeated protests throughout many cities. In early November a new high point was reached when one million people demonstrated in East Berlin on November 4, and several hundred thousand demonstrated in Dresden and Leipzig on November 6. The regime again made concessions, mostly in the form of a new law about travel to the West, but when protesting masses rejected this as being too small of a concession, the regime was forced to enter a fifth stage of major concessions, the most dramatic being the opening of the Berlin Wall on November 9.

The process of demonstrations and regime concessions continued throughout

the following weeks, but the political turning point had clearly been reached at this moment and the further political changes toward a multiparty democracy need not be elaborated here.

Falsifying the Theory: The Chinese Case

A short comparison of our cases to the Chinese case helps to clarify the conditions for the different outcome. Analysis of events leading up to the massacre in Tiananmen Square in June 1989 shows that both protesters and regime actors calculated differently on some key points.

Looking at our schema of protester-regime interaction as outlined in table 1, the Chinese case fits the first and second stages but then begins to differ. The differences in regime responses are especially poignant. In stage three (early May), the regime used a dualistic approach. While giving minor concessions to students in the form of a limited dialogue and allowing freer media coverage of events, repressive controls over workers were tightened by, for example, workers being locked into their enterprises at the time of demonstrations (Walder 1989, 35–37). As a result the bandwagon effect among workers—and at the same time among society at large—was stopped and the outcome of this stage differed from that in Czechoslovakia and East Germany. In the fourth stage, the Chinese authorities declare martial law but are indecisive in carrying it out when confronted by student-led human barriers in Beijing. The initial outcome is a standoff with the numbers of protesters remaining large, but the bandwagon effect remaining on halt. After a few days the numbers of student protesters begin to decrease and are down to 10,000 diehard hungerstrikers in Tiananmen Square when the final stage is reached on June 3rd. The regime's brutal attack on the hungerstrikers effectively ends the mass protests.

In terms of an assurance game, the process works well until the student tipping point has been reached and the first workers have begun to join the protest, but then stops. There were several reasons for this. The first involves the calculus of costs by potential worker protesters. Compared to Eastern Europe, coercive controls at the workplace were more all-encompassing and effective in China (Walder 1989). In China, access to most social benefits, including housing, is tied to the workplace, and due to low labor-turnover workers are dependent for work on their factory. The enterprise bosses actively used this power to deter workers from joining protests and applied other tactics such as locking workers into factories during protests in early May (Walder 1989, 35–37). Past experience also suggested that in confrontations with security forces, workers would be dealt with much more brutally than students. Chinese authorities have a record of being more lenient with student protesters than with worker protesters (Perry 1991). Therefore, Chinese workers had more difficulty obtaining protection from possible costs of protest. In effect, the starting tipping point for the Chinese workers was much higher than for their East European counterparts.

In addition, there was a difference in the predictive calculations of workers. In

China, the workers were not seen as a significant asset in pressing for regime change. In contrast to Czechoslovakia where students appealed to workers to join the protests, Chinese students felt that they could achieve their goals by relying on themselves. The students wanted to be spokesmen for the masses and disregarded workers and peasants (Perry 1991, 134). The disdain of students for workers went so far that at one demonstration they literally linked arms to prevent workers from joining directly in their ranks (Perry 1991, 139; Dittmer 1989, 7). Some workers nevertheless did engage in protests, but as a group they never approached or reached a tipping point. By late May, the regime was arresting labor leaders and engaging in other forms of repression (Perry 1991, 140).

While the Chinese students miscalculated the significance of workers support, another crucial difference to Czechoslovakia and East Germany was that the protesters never challenged the nature of the regime itself. They did not seek an ever-widening popular upsurge because their revolution was self-limiting. Public goods motivations were less encompassing because the public good itself was more narrowly defined. Following Chinese tradition, the students acted as "remonstrators rather than opponents" (Nathan 1989, 17; Perry 1991). The protesters accepted the regime's right of rule and focused on the need to fight corruption and introduce some political reforms. The students wanted a place on the political stage and were seeking patrons within the regime (Perry 1991, 132–33). When the regime elite split and some concessions were made during the midphase of the protests, this reinforced the students' assumption that they could be successful without larger mass activity. As the hunger strikes on Tiananmen Square illustrated, the Chinese students preferred a form of protest that emphasized their moral righteousness and willingness to sacrifice themselves. Even after martial law was introduced, the students stuck to their line of reasoning. In contrast to the effect of similar regime repression in Eastern Europe, this regime action did not turn into a focal event triggering significantly new worker protest and getting closer to their tipping point.

The Chinese case underlies the importance of numerical increase of protest action over time as well as the importance of a dynamic progression toward new group tipping points and the overall societal tipping point. After the student tipping point had been reached in China, and after they had been joined by individual representatives of other social groups, especially by the nascent group of young entrepreneurs (Perry 1991, 136–37), the number of protesters remained large but basically unchanged. By late May, the students even allowed the number of protesters to dwindle, making it possible for a less costly crackdown to occur. In China, intense repression was used at a time when it had the best chance of working, e.g., when the demonstration process was waning.

In China, the progression and interaction of multiple assurance games was stopped, and the same is true for the deassurance games. As in Eastern Europe, the press was the first regime group to tip toward supporting the protesters, but then the process basically stops except for the split in the regime elite. Paradoxically, the support given to the protesters by softliners may have contributed to the stopping of deassurance of the regime as a whole because the hardliners could

target the softliners as enemies to be dealt with in a traditional power-struggle way rather than presenting a factor in deassurance.

In sum, the comparison to the Chinese case shows that outcomes vary depending on the calculus and actions of the regime and protest actors. The Chinese regime's use of repression and concessions was more appropriate for various stages of protest events, and the combining of some initial concessions to students with elaborate repression of workers was especially effective to stop the bandwagon effect among workers and society at large. In Eastern Europe, focal events, public goods motivations, and demonstration effects came together dramatically lowering tipping points in the third stage of the demonstration process helping to link dissidents and students with workers and thus sustaining the snowball dynamic. In China, worker tipping points were raised by regime action, social motivation did not spread throughout the population, student strategies did not take advantage of assembling mechanisms, and regime violence did not act as a focal event because it was not employed until the process was receding.

A significant historical and contextual factor was that the Chinese regime had more legitimacy due to having been established by a homegrown revolution. As noted, the Chinese protesters did not challenge the regime itself, in contrast to protesters in Czechoslovakia and East Germany. There the protester also had an easier time challenging the legitimacy of the regimes, especially so in Czechoslovakia where any lingering legitimacy was destroyed by the invasion of 1968 and the subsequent widespread purge of reformist regime elites.

Conclusions

Many approaches could be taken to analyze the swift overturn of old-style communist regimes in Eastern Europe in the fall of 1989. We focus on how mass protest could arise and how the effectiveness of regime repression was undermined. These are crucial questions because we assume that the inclination to protest was present long before the rebellion, and we therefore have to explain why it was expressed at this particular time and with such forcefulness. We also assume that in spite of pressures for reforming emanating from abroad, ultimately the course of events has to be explained within the political context of each individual East European country. Clearly, the regimes were eager to retain control and were wiling to do so at great cost. Yet when it came to the test, even the Romanian regime's use of shoot-to-kill orders was unable to contain the rebellion.

For the East European regimes, the most shocking aspect was the breakdown of effective deterrence and punishment. For the protesters, the most shocking and exhilarating aspect was the unexpected success of their demonstrations. Our focus has been to explain the dynamics of the rising tide of protest and its links to the decreasing effectiveness of repression. Thinking in terms of an assurance game has emerged as helpful in capturing the pattern of recalculations by protesters. Close attention to the calculations and miscalculations of the regime is also crucial in explaining the outcome at each stage of the progressively larger demonstrations as

well as the end result. Future research should clarify in more detail how regime groups went through a "de-assurance" game.

In the course of the rebellion, the assessments of actors are continually changing. Individual protesters use two main sorts of cues: first, the number of protesters in their own groups, in other social groups, and the rates of observed change and second, the regime's response. Here, the crucial point is that a threat or repressive act is interpreted differently by protesters at the start of the series of demonstrations and at its end. Regimes miscalculated because they looked back to previous experiences for cues and failed to perceive that the use of brute force at the wrong moment backfires.

The evolution of mass protest includes a strategic updating by both sides of the balance between dissent and coercion. Mass protest escalates because a series of progressively larger demonstrations assures many people that very soon the point will be reached where they can avoid punishments and cause meaningful political change. The solution to the assurance game developed here, best seen in East Germany and Czechoslovakia, shows how strategic cues embedded within a heterogeneous society can trigger the movement of larger and larger groups of people toward a tipping point.

Another crucial aspect of the dynamic evolution of the rebellion concerns the focal events when repression itself becomes a target of protest that works to unite the population. This underlies the twofold nature of coercion: it is most effective as the power of preemption, while the use of actual punishment can easily backfire. Outrage about police brutality against peaceful demonstrators expressing legitimate demands backfired on the regimes in East Germany, Czechoslovakia, and Romania. The momentum of demonstrations had reached the tipping point for some groups, and others were coming closer to it. In this context, protesters and potential protesters were not scared off by the use of police force, but to the contrary, rallied around the issue of police brutality and increased the costs for the regime in using repression in the future.

Arrogant confidence in the usefulness of repression most clearly was the downfall of the Romanian dictator. But the neo-Stalinist regimes of East Germany and Czechoslovakia miscalculated as well. Having misused the name of "the people" for decades, the Communist leaders underestimated the real power of the people once it saw the chance to rebel.

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